MP1 Experiment Report

1. Task, Scoring Protocol and Hard Constraints

1.1 Task and scoring protocol

Train a language model from scratch and push WikiText-2 full-test BPB as low as possible, in FP32, reproducible on CPU. Protocol: WikiText-2 raw text with a BPE-2048 tokenizer fitted on the training text only; independent 256-target causal windows that share one boundary token but carry no state; every target scored once, with each split's first token excluded; the BPB denominator is every raw UTF-8 byte of the split; the submission is the test JSON's `bpb` field.

The supplied baseline is full-test BPB **2.1012597** (validation 2.0710831), architecture `d4w128h4` (1,088,256 parameters), 1,200 × 32 × 256 = 9,830,400 training targets (2.7206 epochs); it takes 5.92 s on the reference machine (4-thread Xeon Platinum 8457C, not guaranteed here). With only 3,613,343 tokens at 2.7206 epochs the problem is **data-constrained**: capacity, training volume and regularisation must be balanced jointly. Scaling laws from the 900B-token / 9B-parameter scale, orders of magnitude away, only indicate direction. Our evidence: doubling the training volume again (43.5 → 87 epochs) saturates at **p = 0.7452** (§5.2); a single model at 471.9M targets still sits at ≈1.67 validation.

1.2 Three hard constraints: only one is a real constraint

The shipped artefact was measured on the test split, FP32, 4 threads, **alternating within the same session** with the baseline: CPU scoring time **4.2910×** (MAX of 5 alternating pairs; min/median 3.9523 / 3.9611), peak RAM **2,205.0 MiB**, inference assets **24.938 MiB = 26,149,082 B**.

Only CPU time is a real constraint. Budget-level estimates for assets and RAM (never measured) are superseded by this section's measurement; the qualitative judgement (CPU is the real constraint) agrees, but **the numbers are no longer cited**. The 30 candidate gate decisions corroborate this: every FAIL was on R1; R2 / R3 never neared their ceilings (RAM 2204.7–2230.8 MiB vs 4096; assets 24.915–37.730 MiB vs 64).

R1 is a ratio, and the variance sits in the denominator. Within one validation session the baseline per-target cost spread 21% (14.38–17.44 µs), the ensemble only 4.7%. We read on MAX, because a submission is scored once; the mean would understate the critical value by 5.3×. The same procedure gives 4.6129× on validation, the gap from test coming from the pairing, not the model. The margin boundary (with the failure threshold) is given once only, in §10.1; this section states no other margin figure.

1.3 Compliance red lines and known deviations

All three members have `parent_checkpoints = 'none'`, trained from random initialization, no external corpus or pretrained weights; the only learned asset loaded at inference is `bundle/checkpoint.pt`; all selection on the A half of validation, all reporting on the B half; test was read 6 times (1 frozen score plus 5 alternating timing pairs), all returning the same `1.4574991` under the `--frozen` flag, and took part in no selection, tuning or ranking. The supplied evaluator, model, training scripts, `configs/baseline.json`, `tests/` and all `data/` are byte-identical (14 files); the method was frozen before scoring.

One deviation is disclosed (not a red-line violation): two prose files in the release package do not match their manifest hashes. §4 of `code/README.md` had three blank measurement lines, filled in at the publisher's invitation; the guide's hash is `132409af…` against the manifest's `fa62a767…`; the cause is untraced and the discrepancy predates this work. No file the evaluator loads, nor `data/`, nor the tokenizer, is affected.

2. Experimental Route and Final Method

2.1 Roadmap (Figure 1)

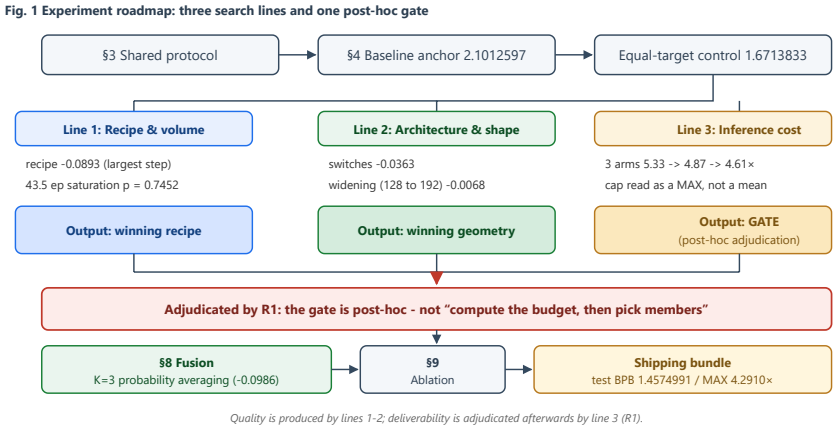


Figure 1, experimental roadmap: the shared conventions (§3), the baseline anchor (§4, 2.1012597), the equal-training-volume control (1.6713833), then three parallel lines — line 1 recipe (−0.0893), line 2 architecture and shape (−0.0363 / −0.0068), line 3 inference cost (5.33 → 4.87 → 4.61×). Line 3's output is not a budget figure but a post-hoc gate; §8 then covers fusion, §9 the ablations, and finally shipment.

2.2 Why this order: the cost line is a post-hoc gate

The equal-target control rules out "it was just trained longer"; the three parallel lines (§5–§7) each answer one separable question. Key structure: **line 3 is a post-hoc gate, not an up-front budget**. The best-quality solutions from lines 1 and 2 repeatedly failed on cost: `b0` (val **1.4342282273**) measured FAIL at 6.5625 / 5.9867; the ungated optimum K=5 (val **1.4185338170**) is not submittable; K=4 / K=5 measured **6.2420× / 6.0603×**, over the gate. The shipped set is limited by R1, not ensemble size; every "best quality" claim must carry its cost.

2.3 Final method at a glance

Three members `d4w192h4` + RoPE + QK-norm[1][2], seeds **35 / 47 / 37**, each seat 157,286,400 targets, loaded with **EMA weights**[6]; shared recipe in §5.3; inference side K = 3 **probability averaging**, loading `agent7_fastens` (`l4_level = 4`); test BPB **1.4574991**, CPU MAX **4.2910×**, RAM **2,205.0 MiB**, assets **24.938 MiB**. Why this is the method: §5–§8.

3. Shared Experimental Conventions

Training volume is `processed_targets = steps × batch_size × 256`. Baseline 1,200 × 32 × 256 = 9,830,400 (2.7206 epochs); shipped members 19,200 × 32 × 256 = **157,286,400** (43.5 epochs). Targets, tokens and windows are three different counts: targets ≠ tokens (the corpus has only 3,613,343 tokens), tokens ≠ windows; `train_tokens` stores a target count, so this report writes **processed training targets**.

For evaluation, validation is 376,599 targets / 1,148,007 B and test is 428,405 targets / 1,292,013 B (each split's token count minus 1). Re-running the same checkpoint is bit-identical, so any BPB difference comes from the weights only. Selection and reporting fix `split_seed = 17`, permuting the validation scored-target indices and halving them: A = 188,299 (selection) / B = 188,300 (reporting); always select on A, report on B; a run's members and their non-EMA twins may not hold two seats at once.

The significance line is **0.0087 BPB** = 2 × pooled se 0.00435 (long-tier three-seed mean, sd 0.00533 / √3 × √2). It is **unpaired**, hence **conservative** for the paired comparisons in this report. Retraining with the same recipe and seed is not bit-reproducible (final BPB differs by +2.0e-5 ~ +6.8e-4), so 0.0087 applies only to multi-seed means, never a single-seed pair; reporting always uses ≥3 seeds as mean ± std. On cost, `Σrel` sums the members' geometric rel; `ratio = Σrel × k / base`, read on MAX; `k_plain = 5.73`, `k_active = 7.04` s/rel; the implementation-side discount is **0.859** (most conservative). The shipped bundle has Σrel = **4.131128** (raw 29.08 s, discounted 24.98 s). Four Σrel-ceiling values appeared (4.92 / 5.9834 / ≈5.142 / 5.5546), **reported in parallel without adjudication**; when conventions conflict, the measured MAX governs.

4. Baseline Reproduction and Equal-Training-Volume Control

Baseline reproduction: official full-test BPB 2.1012597, our value bit-identical (validation 2.0710831). The equal-target control is the denominator for every later Δ: baseline architecture and recipe trained to the same 157,286,400 targets as the shipped members, 3 seeds, same scorer, same split:

	processed targets	validation BPB (3-seed mean)	full-test BPB (3-seed mean)
baseline `d4w128h4`, supplied recipe	157,286,400	**1.6713833**	**1.6952304**
shipped ensemble (157,286,400 per seat)	157,286,400 each	1.4404595	**1.4574991**

Per-seed test readings for the control are 1.6984118 / 1.6962172 / 1.6910621; the ensemble is **0.2336** below the best of them. The alternative "it just trained on 3× more targets" is excluded: 19,200 against 38,400 updates, same architecture and recipe, **p = 0.7452** (saturation); a single model at 471.9M targets still sits at ≈1.67 validation.

5. Line 1 · Recipe and Training Volume

Of the 8 axes, 7 have a measured sweep, 3 seeds per tier, **all pre-registered**: each arm's `abandon_if` fixed its read-out threshold before any numbers existed. Most important: **lr × wd is an interaction term, not the sum of two main effects** (four cells, same architecture, same targets, same seeds, paired control):

Perturbation	Σlr	Σ(lr·wd)	Δ vs anchor	Read
anchor lr1e-3 / wd0.1	1.00×	1.00×	0 (1.6713833)	reference
raise lr only (lr5e-3 / wd0.02, Σ(lr·wd) pinned by back-solving)	5.00×	1.00×	**−0.0004** (paired t = −0.141)	**clean zero**
raise wd only (lr1e-3 / wd0.3)	1.00×	2.99×	+0.0032	**zero**
raise both (lr5e-3 / wd0.1)	5.00×	5.00×	**−0.0367** (t −9.07, 4.21× line)	**real gain**

Conclusion: raising lr and wd together, proportionally (Σ(lr·wd) about 5–6.5×), buys about 0.037–0.039 BPB, not "lr 5e-3" alone. Boundary: that arm has only 3 seeds, df = 2, a "failure to reject the null", not a "proof of zero".

5.1 Results (Table 1)

> Anchor 1.6713833; ✅ = over the line and better, ❌ = over the line and worse, ◯ = within the line; "Δ÷line" = |Δ| / 0.0087. **Table note**: the significance line is an unpaired statistic, conservative for the paired comparisons in this report.

Recipe item	Tier	val BPB	Δ vs anchor	Δ÷line	Over line	Conclusion
anchor	lr1e-3/cos/wd0.1/wu100/drop0/ema0	**1.6713833**	0	—	—	this line's denominator
lr	**5e-3**	**1.6347149**	**−0.0367**	**4.21**	✅	**winner** (2e-3 also over the line, −0.0168)

Recipe item	Tier	val BPB	Δ vs anchor	Δ÷line	Over line	Conclusion
Σ(lr·wd) ladder	1.00× (wd0.02, lr raised only)	1.6709808	**−0.0004**	0.05	◯	**clean zero** (t −0.141)
schedule	**d2z @5e-3 (golden)**	**1.6023611**	**−0.0690**	**7.93**	✅	**this project's largest single-model gain** (t −15.55)
EMA	0.999 (d2z@1e-3 convention)	1.6430812	**−0.0283** (EMA effect −0.0073)	3.25	✅	sub-additive
batch	32 (only tier)	—	—	—	**not measured**	**never swept; not "closed" and not "no gain"**

Tiers that did not make the table: dropout 0.1 / 0.2 gives 1.6826 / 1.7573 (+0.0112 / +0.0860, i.e. 1.29 / 9.88 × line), **monotonically** worse, axis closed; wd = 1.0 is −0.0149 under `decay_scope = matrices` (1.71× line) but collapses to +0.3251 under `all` (attributed to the LN gain/bias group), the narrow-scope wd optimum still unswept; lr = 5e-4 gives +0.0110; Σ(lr·wd) = 15.00× (wd0.3, all) gives a paired +0.0329 (t +4.17). Reported in parallel without adjudication: the EMA effect size is −0.0106 / −0.0073 / ≈0.015~0.017 under three conventions, and this table adopts `d2z`. The recipe alone (lr5e-3 / d2z / wd0.1 / EMA0.999, same architecture, same targets) gives a solo of **1.5820762**, Δ = **−0.0893**, the **largest single step** in the four-step decomposition.

5.2 Training volume (epoch) axis

epoch	steps	targets	val BPB (3 seeds)	adjacent gain / Welch p	verdict
2.72	1,200	9,830,400	2.0742939	—	baseline training volume
5.44	2,400	19,660,800	1.8770846	−0.1972 / not recorded	over the line (22.7× line)
10.9	4,800	39,321,600	1.7472492	−0.1298 / 1.15e-05	significant
21.8	9,600	78,643,200	1.6885748	−0.0587 / 0.00015	significant
43.5	**19,200**	**157,286,400**	**1.6713833**	−0.0172 / 0.0133	significant (t −4.37)
87	38,400	314,572,800	1.6701260	**−0.0013 / 0.7452**	**saturated** (t −0.35)

The per-arm sd ranges from 0.0031 to 0.0053, and **the noise floor varies with step count (a span of about 1.7×), so it must not be carried across regimes**; at 43.5 ep the validation BPB is still falling (t ≈ 4), the slope flattening rather than turning up, so this is not overfitting. Doubling again (76,800 steps, a two-variable change) gives a paired solo $\bar{\Delta}$ of **+0.0782** (6/6 positive), +0.0195 at ensemble-arm level. The short-budget screening trap: the rankings of the 8 geometries at 2.72 ep and 43.5 ep are **exactly inverted** (Spearman ρ = −0.9286, n = 8, 8/8 reversed sign, p ≈ 0.002); acting on the short-tier ranking would ship a model 0.35 BPB worse than the baseline. **Capacity helps at short budgets and hurts at the real one**; the short tier must not be used for architecture / scale decisions.

5.3 Conclusions for this line

Shipped recipe: `lr 0.005` / `d2z` (cosine decay to 10%, warm-down 0.2) / `wd 0.1` / `warmup 100` / `dropout 0` / **EMA 0.999** / FP32 / tied embedding / steps 19,200 / batch 32 / context 256. Closed axes: wd ladder (best arm wins by only 0.0025 < 0.0087, tripping the pre-registered gate); T1.1b decomposition axis (5 of 6 arms worse); dropout (monotonically positive Δ); warmup (flat, non-monotonic, kept at 100); lr frontier > 5e-3 (flat top; 6.5e-3 is not validated); training volume > 43.5 ep (not a methodological contribution). **The batch axis is "not measured", not "closed".**

6. Line 2 · Architecture and Shape

6.1 Shape sweep (Table 2)

The rel and parameter counts come from the cost table, and the best solo validation BPB comes from the 924-run inventory (840 of those runs completed the full 43.5 epochs).

Geometry	Parameters	rel	Best solo val BPB	Verdict
`d4w128h4`	1,088,256	1.000	1.5821	baseline geometry
`d5w128h4`	1,286,528	1.1378	**1.5518**	excellent, steepest in the table (0.046 rel / 0.01 BPB)
`d4w192h4`	2,222,208	**1.3770**	**1.5390** (RoPE+QK-norm variant; plain best 1.5436)	**shipped geometry**
`d4w256h4`	3,749,376	1.8552	1.5463	❌ the evidence does not support "keep widening"
`d8w192h4`	4,001,664	2.2100	**1.5355** (best single model in the table)	❌ very poor value (2.38 rel / 0.01 BPB)

Two further geometries: `d2w256h4` (rel 1.2128, 1.5625) already surpassed, `d3w256h4` (rel 1.5289, 1.5558) a negative gain. The larger geometries `d4w256h8` / `d4w320h4` / `d6w256h8` are also negative gains, or were only trained at lr = 1e-3, but they are **not evidence that "big models don't work"** (full grid in the traceability ledger).

The frontier saturates sharply after rel ≈ 1.4: doubling depth (`d4w192h4` → `d8w192h4`) spends +0.833 rel to buy back only −0.0035, a marginal slope of 2.38, ~50× worse than the 0.046 of the `d5w128h4` segment; a negative-gain segment also appears (`d4w256h4` is 0.478 rel dearer yet 0.0073 worse). **The single-model quality ceiling is ≈ 1.535.**

6.2 Mechanism switches: essentially unchanged parameter count, but not zero cost

The Q4 sweep used same-seed pairing plus a one-sided t read-out (n = 5): 45 arms × 5 seeds = 225 runs; ADOPT 2 / UNDECIDED 19 / REJECT 24, NO-DATA 0.

Switch (arm)	mean(d) BPB	Verdict (t)	Parameter change
RoPE base 10000 alone	−0.0036	UNDECIDED (−0.78)	−32,768
QK-norm alone	−0.0071	UNDECIDED (−1.10)	+512
RoPE + QK-norm + ema999	**−0.0138**	**ADOPT** (t −2.42, p 0.0363)	−32,128
other REJECT arms: RoPE base 100 / no positional encoding / attention temperature	+0.0328 / +0.0919 / +0.0068	REJECT (+19.49 / +7.37 / +2.18)	−32,768 / −32,768 / +4
independent single-variable control for RoPE+QK-norm on the shipped geometry	**−0.0363** (solo 1.5753249 → 1.5390429)	same shape / same seed / same budget / same recipe	see the parameter account

The rows do not share a control: rows 1 / 2 / 4 are against `s3m_d4_golden`, row 3 against `s3m_d5_golden_ema999` (d5w128h4 + EMA999; its −32,128 = −32,768 + 640, where 640 = 5 layers × 2 norms × (g + b) × head_dim 32), each parameter change counted against its own control; all were run on the Q4 budget, not the shipped geometry and budget, so **their effect sizes cannot be added**; −0.0363 is an **independent single-variable control on the shipped geometry**, not the sum of the other rows. Under multiple-comparison correction (BH / FDR, q = 0.01–0.20, all 0 rejections) those 45 arms are null (2.25 expected false positives, exactly 2 ADOPTs observed), the two readings reported in parallel without adjudication. RoPE + QK-norm is adopted on **that control row, not on the Q4 sweep**. The parameter account confirms the switches take effect: 2,222,208 (plain `d4w192h4`) − 49,152 (position table 256 × 192, gone after RoPE) + 768 (4 layers × 2 norms × (g + b) × head_dim 48) = **2,173,824**, bit-identical to every member's `metrics.json`. On CPU cost the mechanism variants are **+31% ~ +53%** dearer at an **essentially unchanged parameter count**, the only group over the "30% recoverable band".

The list trimmed on literature priors (RoPE, QK-Norm, Muon, label smoothing)[1][2][5] falls into three states: RoPE + QK-norm **is supported by measurement and ships** (−0.0363), Muon **was later measured and killed** (§11), label smoothing **has never been run and counts as "not measured"**, not a negative result. **This implementation's QK-norm differs from [2] in one respect**: [2] applies LayerNorm to the queries and keys with centring and bias removed; here it keeps its affine parameters. The same normalisation is also applied **per head** (`head_dim = width / heads`), before the positional rotation.

6.3 Conclusions for this line

The shipped members are not the best solo single models in the whole project (1.5390 > 1.5355); they were chosen for **quality per unit of rel + three seats of the same shape + passing the gate**, not for single-model optimality.

- Ceiling = **1.5355** (`d8w192h4`, rel 2.21): the ensemble is **0.0986 = 11.3× line** better than its own best member; matching it with a single model would need rel ≈ 4.8 (an optimistic extrapolation), while the largest rel measured is 2.61.
- That ceiling **no fair-condition run has yet overturned** (the two largest geometries were only ever trained at lr = 1e-3 project-wide); four "best single-model" values (**1.5355 / 1.5379 / 1.5390429 / 1.552136**) come from different conventions and are **reported in parallel without adjudication**; the body cites only 1.5355 and 1.5390429.

7. Line 3 · Inference Cost and Budget

7.1 The cost model's role and its failure

The cost model is a filter, not an instrument: it screens before submission and flags overruns; ranking errors on the order of 3% have no consequence, and only prediction errors beyond the 30% band are booked. It did fail: parameter count does not predict CPU time (the table below shares one baseline end-to-end reading of 6.057 s):

model	parameters	layers × width	measured scorer seconds
baseline	1,088,256	4 × 128	6.057
`d5w128h4`	1,286,528	5 × 128	**8.258**
`d4w192h4` (plain)	2,222,208	4 × 192	7.893
`d8w192h4`	4,001,664	8 × 192	13.134

"A 128-wide model has 42% fewer parameters than a 192-wide one and is 4.6% slower". This contrast is explained by **the latency term (depth) dominates the FLOP-like account at this scale** (same-source evidence: `d8w192h4` has 1.80× the parameters but only 1.66× the time).

Cases where the prediction misled us: one ensemble predicted at 3.72×, measured at 4.90–5.70×, rejected on that basis; one K=4 bundle was blocked by the prediction and never measured, the after-the-fact measurement giving 5.9036, ✗. The conclusion matches the prediction, but the decision order — a prediction in place of a measurement — was wrong. Folding the member axis into the batch axis gives **68.04 s vs 22.81 s = a 2.98× regression**, **a memory-traffic problem, not a FLOP problem**. The Σrel threshold is no more usable as a gate: a set with Σrel = 4.964 < 5.0 still measured FAIL. Almost all the spread in the ratio sits in the denominator; we read on MAX (a submission is scored once); two disciplines: CPU timing is unusable while a GPU training wave runs (one three-arm reading was voided by CPU contention and re-run on a quiet machine), and the A / B arms must have symmetric memory conditions.

7.2 Three inference-side optimisation tiers (Table 3)

Three arms were run back to back in the same session, on the same frozen checkpoint and a quiet machine, taking the worst of 5 runs:

Tier	Implementation	Worst ratio	BPB	Verdict
control	`student` path, `l4_level = 0`	**5.3277×**	1.4404595	**FAILS** (> 5.0); the unaccelerated bundle does not pass the gate
optimised tier 1	`agent7_fastens`, `l4_level = 1`	**4.8666×**	1.4404595	OK
optimised tier 2 (shipped)	`agent7_fastens`, `l4_level = 4`	**4.6129×** (val) / test MAX **4.2910×**	1.4404595	**OK (shipped)**

It changes only the order in which the arithmetic is combined, not the computation: all three arms give BPB **1.4404595, identical across all 15 repeats**; the load-bearing check is the weight fingerprint — the shipped bundle's per-member fp32 tensor bytes (name-sorted, concatenated) sha256 = `e6defa8b…ae662` / `8000066c…26df9c` / `a034204d…8d7da`, reproduced bit-identically after repacking into a different serialisation container; the bit-exact nll gate `nll_nats = 1146228.1313961744` agrees across all three arms. Without it the same file measures 5.3277× FAIL in the control arm, so this is an R1 **correctness requirement, not an optional optimisation**.

The level 4 increment over level 1 varies in direction across four sessions / conventions (0.8659 / 1.00830 / 0.9991 / level 2 = 0.9228), **reported in parallel without adjudication**. This report's account of the gate boundary **does not depend on level 4**; what is load-bearing is level 1.

7.3 Conclusions for this line

- Conversion rate: taking the most unfavourable observed denominator of 5.641, ratio ≤ 5.0 allows Σrel = 4.92; after a −15% cost-engineering delivery it allows 5.79. Worked out, "**every 10% cut in cost engineering is about 0.5 rel given away for free ≈ half a seat**".
- **The ratio that measurement actually buys back**: three readings of the fastens coefficient, 0.8869 / 0.9052 / 0.8659, a range of **0.865–0.905**; the accounting uses 0.859 (most conservative).
- Residual risk: whether the gate holds depends on the grader's baseline reading, and **the margin boundary (including the failure threshold) is in §10.1**; all 10 measurements spanning two splits were inside the gate. This section gives no other margin figure.

8. Fusion: from a Single Model to a K-Member Ensemble

8.1 Mechanism definition

$\bar{p}(y \mid x_{\le t}) = (1/K)\ \Sigma_k\ p_k(y \mid x_{\le t})$, i.e. `log p̄ = logaddexp_k(log p_k) − log K`.

Why probability averaging, not weight averaging? Weight averaging requires the members to fall in the same loss basin; probability averaging requires only that each be a valid predictor on its own, and it **tolerates active disagreement between members** — "disagreeing with each other" is exactly where the ensemble gain comes from. §9.4 measures why the weight-space version fails here; probability averaging's gain comes from the seat-removal ablation in §9.1 (−0.0986, 11.3× line).

8.2 Candidate pool, pruning and pricing

How the tier-setting search narrowed down: deduplicated 922 → priceable 898 → 277 remain after removing non-activated families → cost-feasibility pruning removed nothing → quality pruning kept 60 / discarded 217 → exhaustive enumeration of **36,050** K ≤ 3 subsets (= C(60,3)+C(60,2)+C(60,1), bit-identical to the script's printout, **enumeration complete**) → 30,654 within budget. "Not found" must be kept apart from "not looked at": what was enumerated is those 36,050 K ≤ 3 subsets; **K ≥ 4 over the same 60-member pool (`C(60,4) = 487,635`) was never enumerated, which falls under "not looked at", not "not there", and is recorded as not measured**. Root cause: **the tool's default enumeration ceiling quietly defined the search space** — when `C(n,k)` exceeds the ceiling the search prints one SKIPPED line and moves on (its greedy table hard-codes `for k in (1,2,3,4)`). Cost-feasibility pruning is a dominance argument (nothing removed from the 277); quality pruning is a heuristic and must be self-verified: the best discarded B-tier solo is 1.5736129, 40th of all 277, and **all top-10 B-tier solo scores are retained**; its limitation: the K = 3 cooperative optimum may lie outside the pool. Pricing goes by `ens_seconds ≈ Σrel × k`; the shipped candidates use `Σrel × 7.04 × 0.859`.

8.3 Member selection: a pre-registered null result

The pre-registered threshold, fixed before the run: the new bundle's half-B BPB must be more than 0.0087 below the **incumbent 1.4431457**, and its cost must pass the gate. Measured, the best within budget on the A half is `s40 + s41 + s35`, `bpb_B = 1.4410330`, **Δ = −0.00211** < 0.0087, **no adoption, incumbent maintained** — **a null result, reported honestly as null**: the 4 new seeds brought no gain beyond noise. Three accompanying facts: (i) the new members are better solo (s40 1.5368796, against the shipped three's 1.5399 / 1.5399 / 1.5390), but swapping them into the triple moves it by only −0.0021, a **very low transfer rate**; (ii) **a better bundle does exist, and what blocks it is R1 rather than quality**: the last row of the Pareto frontier has `bpb_B = 1.4323006` (0.0108 below the incumbent, past the significance line), yet FAILs under both denominators; about 0.0108 BPB was given up deliberately to hold R1; (iii) winner's curse: the incumbent ranks 1194 on the A half and 1247 on the B half out of the 36,050 subsets, a drift of 53 places.

The independent exhaustive search for K = 4 also returns null: neither 114,855 subsets (raw ≤ 31.5 s) nor a second pass over 73,815 feasible combinations (raw ≤ 30.1 s) beat the shipped set; the best compliant k = 4

was 0.0224 and 0.0307 BPB worse. Both ran on the "compliant pool", not the 60-member pool of §8.2 (where K ≥ 4 was never enumerated). Within the gate, the marginal gain from k = 3 → 4 is +0.0224 / +0.0245 / +0.0307 across three sub-sweeps, reported in parallel without adjudication. The null holds only for "the pool enumerated + those two searches".

8.4 The final shipped bundle (Table 4)

Item	Value
Member geometry × K / seeds; parameters / training targets per member	`d4w192h4` + RoPE + QK-norm, all three seats identical (K = 3); seeds **35 / 47 / 37**; **2,173,824** / **157,286,400** (19,200 × 32 × 256), total parameters 6,521,472
Per-member solo validation BPB	1.5398847 (s35) / 1.5398679 (s47) / 1.5390429 (s37)
val BPB / full-test BPB (frozen)	**1.4404595** (half-A 1.4377736 / half-B 1.4431457) / **1.4574991** (exact `1.4574990646`)
CPU ratio (MAX)	test **4.2910×** (3.9523 / 3.9611 / **4.2910**); validation **4.6129×**. This row reports the measured multiples only; **the margin boundary (including the failure threshold) is in §10.1**
Peak RAM / inference assets	**2,205.0 MiB** / **24.938 MiB = 26,149,082 B** (margins and their failure thresholds are in Table 7, §10.1)
Σrel / cost / loading	**4.131128**; raw **29.08 s** / fastens-discounted **24.98 s** / `implementation = 'agent7_fastens', l4_level = 4`; EMA weights; K = 3 probability averaging

Any citation of the cost figures has to carry the qualifiers: `b1_3d4` and `g1` are the same bundle (BPB 1.4404595450, Σrel 4.131128, model 29.08 s, all bit-identical); one "measured FAIL: MAX 5.2273" reading is **without fastens**, and with fastens is 4.5733.

8.5 Quality–cost frontier (Figure 2)

Fig. 2 Quality–cost frontier (validation split)

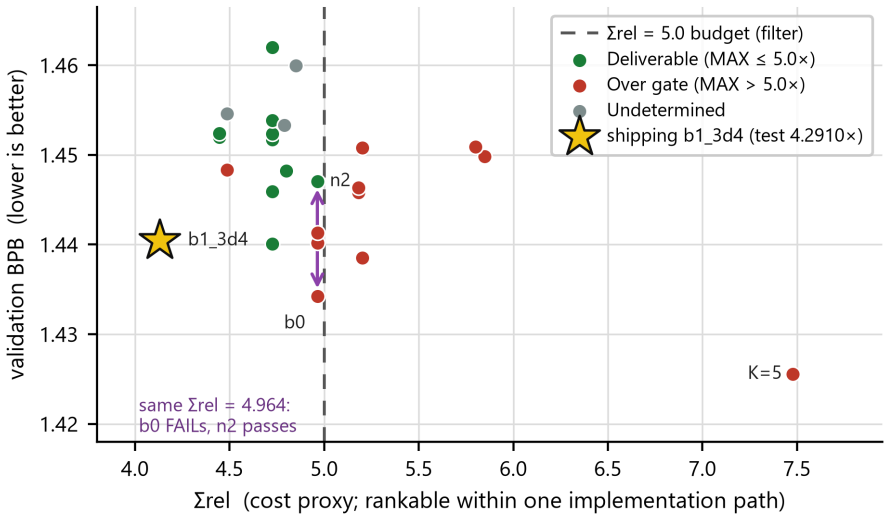


Figure 2, quality–cost frontier. The horizontal axis is the cost proxy **Σrel** (orderable only within one implementation path; across paths multiply by `k_plain = 5.73` or `k_active = 7.04` s/rel), the vertical axis is **val BPB** (lower is better), and the threshold is the **MAX of the measured ratio** (≤ 5.0× shippable, > 5.0× over the gate). The **29 points** come from this section's pricing sheet (32 rows, 3 with no recorded Σrel); green = shippable, red = over the gate, grey = quality not detected. The **star** marks the shipped point; the four points this section discusses — `b1_3d4`, `b0`, `n2` and the K=5 footnote — are labelled in place, and the vertical double arrow marks the two that share Σrel = 4.964 yet land on opposite sides of the gate; the vertical dashed line at Σrel = 5.0 is only a budget filter, **not the real boundary**.

Σrel works only as a filter. `b0` has Σrel **4.964132** < 5.0, yet measures FAIL at **6.5625 / 5.9867**; `n2` has the **same Σrel of 4.964132** yet PASSes at **4.5633** (the lowest ratio observed). The red and green point at the same Σrel is the direct evidence: **the real boundary can only come from measuring bundle by bundle**. Three points must be marked: **`b0`** (val **1.4342282273**, better than the shipped point yet unable to pass the gate); the **K=5 footnote** (val **1.4255519690**, Σrel **7.478716**, over budget and sitting right at the gate, whose seeds for seats 3–5 **were not found in the repository archive, marked * as they stand (recorded as not measured), and must not be filled in**); and **`b1_3d4`** (the shipped point, test MAX **4.2910×**).

On the ungated side there are two more anchors: K = 5 val **1.4185338170** (not submittable); the k sequence (K = 1→5) 1.5398847 / 1.4675860 / 1.4404595 / 1.4263119 / 1.4185338 improves monotonically, but k = 4 / 5 measured 6.2420× / 6.0603×, over the gate. **That sequence is internally inconsistent in convention** (the third entry, 1.4404595, is a full-validation value, whereas half-B k = 3 should be 1.4431457), reported in parallel without adjudication, and **the figure must not mix them**. In one sentence: **"the frontier saturates sharply after rel ≈ 1.4; the whole 0.10 BPB from the best single model at 1.535 to the best ensemble at 1.434 comes from ensembling (variance reduction), not from scale."**

9. Ablations

9.1 Main-mechanism ablation: removing seats (Table 5)

The key mechanism is probability averaging over three members; ablating it means removing seats. **All 7 non-empty subsets** are scored on validation, ordered on A and reported on B:

K	Subset	half-B BPB	Δ vs shipped set	Δ / 0.0087
1	s37 / s35 / s47	1.5413965 / 1.5430833 / 1.5433412	+0.0983 / +0.0999 / +0.1002	**11.3 / 11.5 / 11.5**
2	s35+s37 / s47+s37	1.4696352 / 1.4696400	+0.0265 / +0.0265	**3.0 / 3.0**
2	s35 + s47	1.4708625	+0.0277	**3.2**
3	**s35 + s47 + s37 (shipped set)**	**1.4431457**	**0**	**0**

Every removal crosses the significance line, including the cheapest one (+0.0265, 3.0× line). The ensemble is not a marginal effect and the third seat is not redundant; K = 3 is not yet saturated, but K = 4 cannot pass the cost gate, so 3 is the cap. This table is also the price list for the seat bought to pass the gate: dropping to the best pair costs **+0.0265 BPB**, a **demonstrated loss, not a free hedge**.

9.2 Gain decomposition (Table 6)

The path from 1.6952304 (the equal-target control's full-test) to 1.4574991 mixes four changes, each step holding the others fixed; the last two steps are single-variable (same shape, same seed, same budget, same recipe).

Step	Control variables (what is held fixed)	Δ (val)	Share of the val total 0.2310
start: baseline `d4w128h4`, supplied recipe	—	— (**1.6713833**)	—
+ recipe (lr5e-3, d2z, wd0.1)	same architecture, same targets; 3-seed search optimum = 1.5820762	**−0.0893**	**38.7%**
+ width (128 → 192)	plain architecture, recipe fixed; **same-seed 37 pairing (not the whole-library optimum)**, plain solo 1.5753249	**−0.0068**	**2.9%**
+ RoPE & QK-norm	shape and seed fixed at `d4w192h4` / 37; variant solo 1.5390429	**−0.0363**	**15.7%**
+ ensemble (1 member → 3 seeds)	§9.1	**−0.0986**	**42.7%**
total (sum of the four steps)		**−0.2310**	**100.0%**

The decomposition reconciles: the four-step sum, −0.2310, **≈** the measured total difference of −0.2309 (1.6713833 → 1.4404595; 0.2309238 rounds to 0.2309, while the 0.2310 is four individually rounded terms summed, so they differ by 1e-4). Qualifier: "width" here takes the **same-seed paired value for seed 37** (1.5753249); the plain best **1.5436** for `d4w192h4` in Table 2 is a **whole-library best-solo** convention, **and the two must not be subtracted**; the decomposition reconciles only for the former. Heaviest item: **recipe and ensemble together account for about four-fifths (val 81.3% / test 79.0%), and the recipe half of that is free at inference by construction** (lr / schedule / wd / EMA live in the optimizer and the training side and change no operator executed at inference; the pricing formula `Σrel × k` contains geometry terms only). **This is a structural argument, not a paired measurement**; this study did not run the paired experiment of "changing the recipe only and measuring inference seconds". Second: **mechanism switches are worth more than width** (−0.0363 vs −0.0068), netting −48,384 parameters (parameter ledger in §6.2).

9.3 Diversity-source ablation: seeds vs shapes

The three shipped members differ solo within the significance line (a span of only **0.00084**, 1.5390429 – 1.5398847), statistically indistinguishable; yet the ensemble beats its own best member by **0.0986 = 11.3× line**, and removing any member costs 0.0265–0.0277 at k = 2 and 0.098–0.100 at k = 1. **The gain comes from "being wrong in different ways", not from "differences in individual quality".** Now the direct comparison with a shape-diversified ensemble (a negative result: it is worse):

	previous generation, shape-diversified ensemble (earlier submission 1.4620084; members d4w192h4 + d5w128h4 + d8w192h4)	same shape, different seeds (shipped set b1_3d4; members d4w192h4 × 3)
member solo (full val)	1.5753 / 1.5676 / 1.5937	1.5399 / 1.5399 / 1.5390
ensemble K = 3 (full val / half-B)	**1.4620084** / 1.4649171	**1.4404595** / **1.4431457**
Σrel → seconds	**4.7249** (× `k_plain` 5.73) → 27.07 s	4.1311 (× `k_active` 7.04) → 29.08 s

Shape diversification spreads members more widely, but **the ensemble is worse**: the same-shape seed ensemble is better by **+0.0215 (2.5× line)**. The two sides' k differ (left `k_plain = 5.73`, right `k_active = 7.04`), so the "only 2.01 s dearer" comparison **crosses implementation paths** and is an order-of-magnitude reference only. Two qualifiers: (i) the comparison is **between shapes**, the mechanism switches held constant across all three; (ii) **a three-member ablation cannot separate "seed diversity" from "more parameters"**; the "training duration" reading is excluded by the saturation control in §4 (p = 0.7452), not by this ablation.

9.4 Reverse ablation: why weight merging does not work

Merging the three members into one set of weights would cost 1× instead of 3× — the only mechanism that could buy back cost headroom without losing ensemble width. First the instrument is excluded: the symmetry group is solved exactly (residual-stream permutation $\in S_{128}$, per-layer MLP hidden permutation $\in S_{512}$, head permutation and within-head permutation)[4], and the implementation recovers **2704 / 2704 (100.00%)** on a known answer, `max|A − align(B)| = 0`. The barrier[7] survives alignment (local lab: `d4w256h4_s17/18/19`, same architecture and recipe, differing only in seed, single models all 1.8139): **naive average merging 3.7570, barrier = +1.9431**; **aligned merging 3.7407, barrier = +1.9268**, so alignment recovers only 0.0163 BPB.

Two independent chains reproduce this. (i) Pool-member side: a single member `w4_s36_only` **1.6192**, the naive average of three same-family seeds `w4_soup3` **3.4977** (collapse of **+1.8785**); qualifier: a soup is a single predictor competing against a single model, not an ensemble, so even without collapse it could not reach 1.44. **`w8_soup3` was never scored and is recorded as "not measured", not as a failure.** (ii) P2-14 (pre-registered, with CI / MDE, n = 39): cross-seed endpoint average minus own endpoint = **+1.8206** (CI ±0.0501, MDE 0.0712, past the MDE), the cluster level past it too; `SWA4 − delivered-form EMA = +0.0040` (not detected, sign unfavourable); the regression `SWA_gain ≈ 0.8076 × EMA_gain` (R^2 = 0.8433), so the two harvest the same end-of-trajectory variance while EMA is stronger. Instrument limitation: the delivered family has no intermediate ckpt, so the P2-14 reading is a same-schedule-family proxy surface, not the shipped artefact. Top1 match rate in two independent diagnostics: weight matching `pi` **0.406**, activation matching `pi` **0.449**; random baseline 0.0039 / 0.001, CONTROL = 1.000, so the matcher is finding real structure, but both channels are wrong by about 55%–59%, and `pi` top1 falls monotonically from 0.449 at the embedding output to 0.254 at the last layer. **This is not "the same set of features relabelled", it is two different solutions.** Note that **0.406 / 0.449 are match rates, not the relative size of the barrier**; the barrier is always absolute BPB, **+1.9431 (after alignment +1.9268)**.

9.5 What the ablations establish and what they do not

(i) Effect sizes measured. Removing seats crosses the line at every tier (cheapest k = 3→2 is +0.0265 = 3.0× line, dearest k = 3→1 is +0.098~+0.100 = 11.3~11.5× line); shape diversification is **worse by +0.0215 = 2.5× line**; the decomposition closes at −0.2310. The paired bootstrap 95% CI per subset of the shipped set **was not found in the repository**.

(ii) Not established either way (which is not the same as the opposite being established). The alternative path of a larger single model has never been closed: in the priced geometry grid, every shape passes all three gates on its own, and even the worst, `d6w256h8`, reaches only 2.9158× (58.3% of R1) and 20.33 MiB (31.8% of R3); the one width step actually run (128 → 192) is worth only −0.0068. Training longer, the inseparability of seed diversity from parameter count, and the cost model's inability to gate are likewise not established either way.

(iii) Not to be extrapolated. The conclusion holds at ~2M–5M parameters, 4–8 layers, width 128–256, BPE-2048, on 3,613,343 tokens, at 43.5 epochs (a severely over-trained regime); all wall-clock multiples were measured on one machine, while final scoring happens on the grader's machine (the same baseline reads 5.92 s on the reference machine and 8.3929 s here, a factor of 1.42).

10. Resource Compliance, Cost and AI-Use Disclosure

10.1 The three hard constraints as measured (Table 7)

Constraint	Ceiling	Measured (test)	Measured (validation)	Margin
CPU ratio (MAX, 5 alternating pairs)	≤5×	**4.2910×** (3.9523 / 3.9611 / **4.2910**)	**4.6129×**	test **14.2%**; val 7.7%
Peak RAM	≤4 GiB	**2,205.0 MiB**	2,204.9 MiB	**46%**
Inference assets (uncompressed)	≤64 MiB	**24.938 MiB** = 26,149,082 B	—	**61%**

> **Table note**: the full meaning of the "margin" column (including the failure thresholds) is in the margin-boundary paragraph at the end of this section (a margin is not a guarantee). Elsewhere a reference to a margin only points here and does not restate the figures.

Measurement procedure: baseline and ensemble **alternate within the same repeat**, 5 alternating pairs, FP32, 4 threads, the official `evaluate.py`, the `--frozen` flag; 10/10 readings return the same BPB, so what is repeated is time, not score. RAM comes from external RSS measurement, not from `peak_allocated_gb` in the evaluate.py JSON (a CUDA counter, reading 0.0 on CPU).

The honest boundary of the margin (the only full statement in this report): passing the gate requires the grader's baseline to be **≥ 28.616 / 5.0 = 5.72 s** (official reference value 5.92 s); the fastest baseline reading in this session is **5.2232 s**, and under that denominator the test numerator of 28.616 s would read **5.48×, over the gate**. Pairing it with the validation ensemble of 25.277 s gives 4.839× (a margin of only 3.2%), but that is a **cross-session worst case, not a measurement**. The pre-registered mitigation and its price: buy margin with the ensemble size k; dropping to the best two seats costs **+0.0265 BPB = 3.0× line**, a loss measured rather than a free hedge. **This is a margin statement, not a guarantee.**

10.2 Training and search cost

Item	Value
total runs / total GPU hours / total targets	**1,008 runs** (those with metrics.json) / **400.2 GPU·h** / **143.93 G**
cost of the shipped members themselves	**7,280.7 s = 2.02 GPU·h** (per member 2,426.5 / 2,427.6 / 2,426.6 s)

Item	Value
search overhead multiple / CPU-side evaluation and search (est.)	**398.2 GPU·h ≈ 197× the shipped members**, members being 0.51% of total compute / **8–12 CPU·h**

On the principle that "reusing a checkpoint does not erase its training cost", this report gives the full **400.2 GPU·h**. All 1,008 runs record `device = cuda:0`; the members have `parent_checkpoints = 'none'`.

10.3 test usage

test was read 6 times in total: 1 for the frozen score, 5 within the alternating timing pairs of §10.1; all six returned the same `1.4574991` under the `--frozen` flag. **Did it take part in any selection: it did not.** All selection was on the A / B halves of validation; the member pool, the subsets and the pre-registered thresholds were finalised before any test read.

10.4 AI-use disclosure

This work used substantial AI assistance: AI helped draft and iterate the search, enumeration, packaging and measurement tooling, design and run the recipe and architecture searches, write `agent7_fastens.py`, and draft this report. **This includes code the evaluator loads**: the shipped checkpoint's `implementation` field points at `agent7_fastens`, on the scoring path. AI did not write the supplied `model.py` / `evaluate.py` / `common.py`, and did not modify the data or tokenizer; all three members were trained from random initialization. Before shipment we verified that the module does not change the score: the three arms, run on the same checkpoint, all return `bpb 1.4404595` on validation.

11. Negative Results and Boundaries (Table 8)

> Discipline: each row = **verdict + evidence + closing action**; no "the trend is right" and no "it seems worse". Anything that was never run is marked **not measured**, not written as a failure.

#	Direction	Conclusion (verdict)	Closing action
1	cross-seed weight averaging / model soup[3]	**Failure**: naive cross-seed averaging gives val **3.4977** (members solo **1.6192**, collapse **+1.8785**); lab barrier **+1.9431 / +1.9268 aligned**; symmetry group solved, two diagnostics still top1 only 0.406 / 0.449	weight-space merging **closed**; replaced by probability averaging (§8.1)
1b	`w8_soup3`	**not measured** (no reading anywhere in the repository)	the report must write **not measured**, not a failure
2	cost model (Σrel / parameter-count proxy)	**Failure**: 42% fewer parameters yet 4.6% slower; errors +16% (candidate sets), +19% (single model), +31%~+53% (mechanism variants; another convention +12%~+20%, reported in parallel without adjudication); one ensemble predicted 3.72×, measured 4.90–5.70×, rejected on that basis	demoted to a screener / pruner; resource-gate decisions must be taken on end-to-end measurement
3	exhaustive member selection	**null**: exhaustive enumeration of 36,050 K ≤ 3 subsets over the 60-member pool, best selected on A gives **Δ = −0.00211** < 0.0087 ⇒ incumbent maintained	reported honestly as null; the pruning self-verification record is disclosed alongside
5	the four inference-side speed-up levers	**all no gain or backwards**: batching a 2.98× regression (memory traffic); `mallopt` 1.02093 with peak RSS +165.7 MiB; head-chunked parsing upper bound 3.8279 s = 0.4448× of the 8.6067 s requirement; the de-variant `P/S = 0.8588` triggering the criterion	all four rolled back; shipment keeps the plain K serial passes. Head chunking can only be reported as "**the evidence does not support this direction**", not as "the opposite has been established"
7	**Muon optimizer**[5]	**killed: worse, not "not detected"**: accum 1 (= the delivered operating point) Δ = **+0.1399** (CI ±0.0455, MDE 0.0523, n = 2); accum 16 Δ = +0.2210	the pre-registered criterion triggers ⇒ closed
7b	**Label smoothing**	**not measured (never run)**: the code is ready, but there is no result reading anywhere in the repository	the report must write "**not measured**", not a negative result
12	short budget (2.72 epoch) as an architecture screening proxy	**inverted, failure**: Spearman ρ = −0.9286 across the two tiers for 8 architectures (n = 8, 8/8 reversed sign, p ≈ 0.002); selecting on the short-tier ranking picks the worst performer at the long tier	the proxy is closed; all architecture conclusions use 43.5 epoch only
14	**n-gram count fusion**	**adopted on quality → killed at delivery**: adopted 09-22 (half-B −0.0089, ratio 4.5502); on 09-23 the pre-registered criterion "pessimistic envelope margin < 3% ⇒ kill" triggered, measured −0.93%, the pre-shipment reading falling on the FAIL side	removed from the delivery tree; its test reading deliberately not taken
15	shape diversification (buying diversity with shape)	**firm negative result**: **worse by +0.0215 than the same-shape seed ensemble (2.5× the significance line)**	the three seats stay same-shape, different-seed

Other closed directions: training side, self-distillation (Δ **+1.1087**) and training longer, 76,800 steps (paired Δ̄ **+0.0782**, 6/6 positive); architecture side, bigram hash / SwiGLU / mixer replacement (+0.004 ~ +0.522), cross-layer parameter sharing (+0.055 ~ +0.081, all REJECT under 5-seed pairing), mechanism diversity (205 qualifying members, the best 25 within a 0.0034 BPB band); inference side, temperature / member reweighting (peak c = 1.04 worth only −0.00142, −0.0003 after fitting the weights). One class needs its own note: **changing the tokenizer / changing the data (deduplication, curriculum learning, pruning) / external or pretrained weights / MoE / structured sparsity are not applicable (violating the hard constraints or data-**

constrained), not "ineffective"; nor may this be written as "deduplication is ineffective for BPB" or "MoE is ineffective".

12. Reproduction and Delivery Checklist

One-command scoring (run from the `code/` directory):

```
python evaluate.py --checkpoint ../bundle/checkpoint.pt \
    --device cpu --precision fp32 --threads 4 --split test --output peer-test.json
```

Field	Expected value
`bpb`	**1.4574991** (exact `1.4574990646`)
`targets`	**428405**
`utf8_bytes`	**1292013**

(`--split validation` expects `bpb = 1.4404595`.) Artefact identity hashes: bundle `checkpoint.pt = 94b5af1ea5282bf86ec6543cacb3bd90138895b7516f626f8cebfac31af03259` (26,149,082 B) | `agent7_fastens.py = de64fa1e33cb8504840a3d8f6a50a8922378e932eb92da2ce892896496691de4` | `student.py = af8856febd7b3bc31d60753d3970a10eef42a4266e98767512195cae08c66dee` (also the `implementation_sha256` recorded in each member's `metrics.json`). The loading path is `agent7_fastens.py → student.py → model.py (+ data/tokenizer)`; the full installation, training and artefact checklist lives in the repository README (the retraining command is in its §3), and supplied-file consistency is in §1.3.

13. Critical Analysis

The takeaway is not the number "1.4575" but four caveats. **The shipped set is the optimum under the cost constraint, not the quality optimum**: ungated, K = 5 reaches val 1.4185; to hold R1 we gave up about 0.0108 BPB deliberately (§8.3). **The bulk of the gain is not in the architecture**: in the four-step decomposition, recipe (−0.0893) and ensemble (−0.0986) together account for about four-fifths, the recipe half of that being free at inference by construction, while shape and mechanism switches together account for only about one-fifth; this runs against the intuition that "a bigger model would win", but follows from the task's 3.6M-token corpus. **Three alternative paths are still open**: a larger single model (not established either way), K ≥ 4 combinations over the 60-member pool (never enumerated), and the batch and label smoothing axes (never measured); writing any of them as "established against" or as "failed" is wrong. Last: the CPU gate is not robust here, it depends on the grader's baseline reading, and **the margin boundary (including the failure threshold) is in §10.1**; **this is a boundary statement, not a guarantee.**

14. References

This report reuses the following published work (bracketed numbers match the in-text markers).

[1] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen and Y. Liu, "RoFormer: Enhanced Transformer with Rotary Position Embedding", arXiv:2104.09864, 2021 (published in Neurocomputing, 2024).

[2] M. Dehghani, J. Djolonga, B. Mustafa et al., "Scaling Vision Transformers to 22 Billion Parameters", ICML 2023 (PMLR 202), arXiv:2302.05442.

[3] M. Wortsman, G. Ilharco, S. Y. Gadre et al., "Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time", ICML 2022 (PMLR 162: 23965-23998), arXiv:2203.05482.

[4] S. K. Ainsworth, J. Hayase and S. Srinivasa, "Git Re-Basin: Merging Models modulo Permutation Symmetries", ICLR 2023, arXiv:2209.04836.

[5] K. Jordan, Y. Jin, V. Boza, J. You, F. Cesista, L. Newhouse and J. Bernstein, "Muon: An optimizer for hidden layers in neural networks", 2024, https://kellerjordan.github.io/posts/muon/; for large-scale validation see J. Liu, J. Su, X. Yao et al., "Muon is Scalable for LLM Training", arXiv:2502.16982, 2025.

[6] B. T. Polyak and A. B. Juditsky, "Acceleration of Stochastic Approximation by Averaging", SIAM Journal on Control and Optimization, 30(4): 838-855, 1992, DOI 10.1137/0330046.

[7] T. Garipov, P. Izmailov, D. Podoprikhin, D. Vetrov and A. G. Wilson, "Loss Surfaces, Mode Connectivity, and Fast Ensembling of DNNs", NeurIPS 2018, arXiv:1802.10026.